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OVERVIEW

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Understanding synergies and tradeoffs between forests, water, and climate change

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Abstract

Understanding synergies and trade-offs between forests, water, and climate change is warranted for designing effective policies and strategies for managing water and forests, which are essential for sustenance, ecological proliferation, and economic development. Forests are considered global storehouse of resources, functioning as ecosystem service providers, such as recyclers of terrestrial water to maintain quality and quantity of water but are constantly regulated by climatic parameters. These interlinkages are further complicated by the highly debated role of forests in water regulation and consumption, anthropogenic changes in land use, changing climatic patterns and their subsequent impacts on the hydrological cycle. However, policy and planning for natural resource management seldom consider the interrelationships between forest, water, and climate change due to lack of consensus, misrepresentations and difficult conversions of the complicated interactions to policy. We review and discuss the existing research on these interrelationships with different approaches using a range of hydrological, climatic, and land use indicators. We further suggest incorporating long-term data for forest, water, and climate into conceptual, statistical, and stochastic models may yield better projections with fewer uncertainties rather than those focusing on linear interactions between paired components. Thus, there is a need for exploring these interactions holistically rather than in silos from the perspective of natural resource management particularly in developing nations such as India that have a pressing need to develop new and synergize existing strategies for sustainable management of forest and water under changing climatic variables.

This article is categorized under:

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adaptive strategies, forest management, India, water management

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1 | INTRODUCTION

Climate change is being accelerated by anthropogenic emissions of green house gases (GHGs), having severe implications globally for economies, societies as well as the environment (Field et al., 2014). Although it is difficult to define the precise nature of climate change impacts at micro-scales, there is a widespread consensus on key macro-level impacts backed by climate models (McCarthy et al., 2011; van Aalst, 2006). Developing countries, in particular, are highly vulnerable to climate risk due to their lesser adaptive capacity and higher dependence on natural resources (Adger et al., 2018). In addition, 1.6 billion people worldwide are largely dependent on forest resources and linked services for their livelihoods, of which 60 million are indigenous and tribal people (Chao, 2012). The livelihoods of the latter are directly vulnerable to changes in type and extent of forests for a variety of needs such as sourcing food, fodder, timber, climate, and water-related services (Reid et al., 2005). Moreover, changing forest cover has serious implications at ecological, social, and economic levels; resulting in the dilution of forest benefits, provisioning, regulating, supporting, and cultural ecosystem services (Anderson et al., 2011; Bates et al., 2008; Bergkamp et al., 2003; Ellison et al., 2019; Reid et al., 2005). One such underrated benefit that is significantly impacted by changes in forest cover is water regulation (Ellison et al., 2017; Li et al., 2017; Zhang et al., 2017). Forests hold immense influence over water resources at multiple levels and in multiple ways, highlighted by the overlap of severely water-scarce regions with forest-scarce regions (Mekonnen & Hoekstra, 2016; Rockström et al., 2014). Historically, ecohydrological studies focused on the impact of forest conversions on hydrological processes, have also highlighted the role of forests in water regulation (Andréassian, 2004). Although water is a renewable resource covering 70% of the earth's surface area, of which 97% is saltwater, 2.5% is freshwater (stored as polar ice or as groundwater), and only a limited amount (0.5% of the total volume of water on Earth; Oki & Kanae, 2006) is extractable for meeting the domestic, agricultural and industrial demands (Foley et al., 2011; Rockström et al., 2014). This limited accessible water continuously moves through different phases of the hydrological cycle (atmosphere, land surface, and subsurface) via evaporation, evapotranspiration, precipitation, surface water, runoff, groundwater, and groundwater flow (Healy et al., 2007; Kothawale et al., 2010; Raghunath, 2006).

Forests and water resources face additional pressure of an exponentially increasing population, unsustainable extraction of resources, pollution while also being adversely impacted by climate change. It has been documented that even moderate climatic anomalies may result in serious alterations of forest ecosystems (Edenhofer, 2015) in the form of forest composition shift; changes in biomes, structure, function, and species; forest-diebacks and pest attacks (Allen, 2009; Boisvenue & Running, 2006; Bonan, 2008; Chen et al., 2011; Devi et al., 2018; Hanewinkel et al., 2010; Hansen et al., 2001; Patasaraiya et al., 2021). The increased rate and uncertainty of climatic variability (Thornton et al., 2014), also alters the relationship between forests and water flow (Caldwell et al., 2012) and the "climate-regulatory function of forests" necessary for a sustained water regime and availability (Ellison et al., 2012). Significant perturbations in climatic patterns and water cycles in the form of erratic rainfall and increasing temperatures have been observed with a considerable increase in extreme weather events further influencing both regional and global water availability (Bates et al., 2008; Rajeevan et al., 2008; Trenberth, 2011). The climate crisis has worsened water scarcity with speculations of 1.8 billion people at risk of living in absolute water scarcity and two-thirds of the world's population may experience water-stressed conditions by 2025 (FAO, 2020). Even though the provision of clean water is one of the most basic ecosystem service necessary for life on earth, the growing demand for water caused by an increasing human population, combined with the adverse effects of climate change, is creating unprecedented challenges for sustainable development. The inclusion of water (SDG 6: Ensure availability and sustainable management of water and sanitation for all), forests (SDG 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss) and climate (SDG 13: Take urgent action to combat climate change and its impacts) as separate Sustainable Development Goals also highlights the most critical area for conservation of natural resources in the context of the emerging climate crisis and associated risks (UN, 2015). Although there are dedicated goals for forest, water, and climate action interlinked with different goals, the nuances of the interactions are only fleetingly mentioned in the targets (Wierik et al., 2020). Therefore, a synergistic approach is required to undertake the challenges of water security and climate variability (Smajgl et al., 2016). Such initiatives would help prioritize the discussion on resource conservation and proper integration of climate policies with water and forest management.

Interlinkages between forest and climate have been well established and it is widely accepted that forests play a vital biophysical role along with the climate in regulating the hydrological cycle. Alterations in any of the "key forest-water and land-atmosphere interactions" (Ellison et al., 2017; Ellison & Speranza, 2020) may have cascading implications on

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their overall network. Hence, a rapidly changing climate underlines the need to assess the dynamic relationships between forest and water in a new context. There have been several studies on understanding the impact of climate change on forests (Allen, 2009; Devi et al., 2018; Patasaraiya et al., 2021; Ravindranath et al., 2006), agriculture (Ahmad et al., 2011; Mendelsohn, 2008), biodiversity (Lovejoy & Hannah, 2005; Pawson et al., 2013), water (Freas et al., 2008; Kumar, 2012), and land-use (Booij et al., 2019; Khare et al., 2015). Although climate change impacts on forests and water separately have been studied intensively, the inter-relationship between the three have been less explored, especially in the Indian subcontinent. A shift toward an integrated cross-sectoral approach is pivotal for improving policy design, implementation and increasing resource efficiency through natural resource (forest and water) management (Govt. of Japan, 2002). Many scientists have stressed the need to understand and study the convergence of inter-related natural phenomena for the water cycle due to its systemic and cross-sectional nature linked to both natural components such as soils, vegetation, climate, and social components like politics, economy, and law (Ellison et al., 2017; Furniss et al., 2010; Vose et al., 2011; Wierik et al., 2020). Therefore, this review attempts to provide a literature base by examining our current understanding of how forests as a land use fit in the hydrological cycle, the impact of climate change on forested watersheds, and subsequent identification of trade-offs and benefits from these interactions. We also highlight how managing forest, water, and environment is important for achieving sustainable development goals. Through this review, we are trying to validate the shift in the partitioning along the hydrological cycle in context to changing land use and climate variables, the need for an interdisciplinary approach to address this subject and policy formulation pertaining to these interactions, especially for developing nations like India (Box 1).

BOX 1 Indian scenario

Developing economies like India with lower per capita income, social inequality, and significant portion of their population reliant on natural resources and agriculture, rendering them highly vulnerable to changing climatic regime (IPCC, 2014). Forests in India (covering 21.67% of geographical area; ISFR, 2019) are one of the primary sources of food, fodder, timber, non-timber forest products (NTFP), and livelihood for 275 million rural and 88 million tribal people (Reid et al., 2005; Sharma & Chauhan, 2011), which are increasingly under pressure due to over-exploitation, deforestation and climate change. Water being another resource responsible for economic, ecological and social stability has become a consumptive commodity as India had its largest freshwater withdrawals ever (760 billion cubic meters per year) in 2014. According to the UN AQUASTAT dataset, the developing countries used more water for agricultural purposes as the demand in India almost doubled from 1975 to 2010 due to population explosion and increased food production making it the world's largest water consumer (700 billion m^3 per year; Ritchie & Roser, 2017). With water as a limiting factor for survival and growth (GOI, 2012); the Indian scenario seems grim, with a large part of the country suffering from high to extremely high water-stressed conditions (54%) and a predicted 50% fall below demand in the national supply by 2030 (Gassert et al., 2013; WRI, 2015). Furthermore, the availability of water for human use is highly sensitive to the changing climate variables, land-use change, and invasive species, all of which affects the quantity as well as the quality of water, which could potentially debilitate the overall growth of the nation (Allen, 2009; Ellison et al., 2005; Field et al., 2014; Vose et al., 2011). This is especially true for India, the fifth most climate-affected country with the highest death toll and 0.36% losses per unit GDP due to climate change (Eckstein et al., 2019). The warming of the Indian Ocean at a higher rate than other oceans points to increased heat waves, heavy rainfall and flooding (IPCC, 2021). Although India's commitment to sustainable forest governance and management is reflected in the National Forest Policy, 1988, the Indian Forest Act, 1927, the Forest (Conservation) Act, 1980 and the Wildlife (Protection) Act, 1972, the National Mission for a Green India, and the Environment Protection Act 1986; and the management of water resources through the National Water Policy, National Water Mission, National Mission for Clean Ganga, National River Conservation Plan, Water (Prevention and Control of Pollution) Act, 1974, Water Cess Act, 1977; and climate action through National Action Plan for Climate Change, and INDC's; the interaction between forest-water-climate change is missing from the policy discussion and formulation. Therefore, it is essential to explore the complicated relationship between climate change, forest and water, and their role in influencing water availability at national and sub-national levels.

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2 | FOREST AND WATER

Forests resources have gained attention for supporting varied needs of fuel, food, fodder, timber, climate (carbon sequestration, and storage), and water-related services (Bonan, 2008; Creed et al., 2016; Crowther et al., 2014). Forests are acknowledged for their role in regulating water ecosystem services (Ellison et al., 2017; Grizzetti et al., 2016; Kornatowska & Sienkiewicz, 2018) but are relatively underappreciated compared to other services in policy formulation (Wierik et al., 2020). We are focusing on one of the crucial services provided by forests that is, Water regulation. The following sections give an overview of the importance and role of forests in the water cycle, different theories, and the effect of changing land use on forest–water interactions.

2.1 | Biophysical relationship

Forests and trees (structure and function) are intricately connected to the hydrological cycle, crucially driving it both above and below ground (Bonell, 2002; Bruijnzeel, 2004; Calder, 2007). The primary facilitators of tree and water interactions are leaves, stems, roots, and soil which consume and produce water via biophysical processes like transpiration, interception, evaporation, infiltration, and groundwater recharge (D'Odorico et al., 2010; van Noordwijk et al., 2015, 2018). Trees control discharge and runoff by accumulating rainfall/snowfall through canopy interception which helps in the regulation of groundwater recharge, erosion, and streamflow (Duan et al., 2017; Li et al., 2017; Zhou et al., 2011). Hence, forests also contribute toward flood/drought mitigation while indirectly supporting aquatic habitats for biodiversity. These functions constantly undergo natural fluctuations mediated by forests and forest soils, and anthropogenic changes. Tree phenology, age, root patterns and depth, wood density, Leaf Area Index (LAI), forest type, and climatic condition also have a strong influence on ecohydrological interactions (Gebhardt et al., 2014; González de Andrés et al., 2018; Larjavaara & Muller-Landau, 2010; Wright et al., 2017). For example, trees with larger leaf basal areas induce higher rainfall interception; mature root systems of forests contribute toward higher soil moisture extraction and higher transpiration than other vegetative assemblages (Bruijnzeel, 2004; Ice & Stednick, 2004; Neary et al., 2009).

Forests also determine water quality by acting as a natural water treatment unit (Pires, 2004) with forest soils dubbed as the most hydrologically relevant part of forests for water quantity, quality, and stream habitat. The biophysicochemical properties of forest soils facilitate nutrient uptake, cycling, contamination sorption, and thereby moderate stream hydrology providing an adequate source of good quality water (Neary et al., 2009). The cumulative impacts of the above are reflected in the fact that major cities across the globe such as New York, Mumbai, and Bogota are dependent on forested watersheds for their daily domestic, agricultural, industrial, and recreational needs (Creed et al., 2016; Dudley & Stolton, 2003; Ellison et al., 2017; Furniss et al., 2010; Mueller et al., 2013). Forests are essentially "nature-based solutions for water" (Ellison & Speranza, 2020) functioning through climate and water-related services to provide clean and stable water supply for consumption, precipitation recycling, atmospheric cooling, water purification, aquatic habitat, and groundwater recharge (Ellison et al., 2017).

2.2 | Hydro-ecological correlations

Eco-hydrological relationships and their consequential effects on water quantity and quality with respect to changing forest cover have been assessed worldwide using experimental and paired catchment treatments, advanced remote sensing and geographical information system, different statistical, empirical techniques and models (Bosch & Hewlett, 1982; Brown et al., 2005; Moore & Wondzell, 2005; Zhang et al., 2008). Paired catchment studies focusing on the impact of forest transitions on streamflow and water yield reinforce the idea that reduced forest cover leads to increased water yields whereas reforestation decreases water yield (Andréassian, 2004; Brown et al., 2005; Bruijnzeel, 2004; Hibbert, 1965; Moore & Wondzell, 2005; Sahin & Hall, 1996; van Dijk et al., 2012; Zhang et al., 2008). This increase in annual runoff due to forest loss is attributed to a reduction in interception, evapotranspiration, and higher precipitation than potential evaporation, generating more abundant water flow (Andréassian, 2004; Stednick, 2008). However, inconsistency in the results due to different factors (size, scale, forest age, afforestation, and reforestation) accounts for varied responses of runoff to forest cover change (Stednick, 2008).

A global review on hydrological responses to forest cover changes has made comparisons between small and large watersheds based on impacts on different spatial scales, climate, forest types, and hydrological regimes (Zhang

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et al., 2017). All comparisons converge by correlating increased runoff with decreased forest cover with strong statistical significance even at different scales. However, there are discrepancies in quantifying the impact of forest cover gain at varying scales. Overall, hydrological regimes in water-limited watersheds are more sensitive to changes in forest cover as compared to energy-limited watersheds and was found to be consistent across varying spatial scales. Forest type, soil, topography, climate, and landscape pattern are major factors for significant variations in the hydrological response to forest change in small watersheds (Moore & Wondzell, 2005; Zhang & Wei, 2014). In particular, forest type is a significant variable in influencing annual runoff in response to forest cover change across small watersheds, as small mixed forest or large snow-dominated forests are more hydrologically resilient to forest cover change (Zhang et al., 2017). Small mixed forests have higher resilience and stable water yield in response to changing climate (Brown et al., 2005; Creed et al., 2014; Zhou et al., 2011) as different species with phenological variations (canopy, rooting depth, fruiting, flowering, and leaf patterns) minimize competitive interactions for water usage and use water more efficiently (González de Andrés et al., 2017; González de Andrés et al., 2018). The type of watershed also has significant influence on this interaction as logging in snow-dominated hydrological systems had more pronounced impacts on annual runoff than in rain-dominated hydrological systems (Moore & Wondzell, 2005).

Shifts in species composition and structural changes in forests may also strongly influence water yield and flow regimes. A study in Tahe watershed, China observed that as young larch trees and secondary birch forests replaced old-growth larch forests, there was a reduction in mean annual streamflow along with an increase in the magnitude of high/low flows (Yu et al., 2019). On a similar note, Singh and Mishra (2014) studied water quality amongst four watersheds with different forest profiles of old forests (primary forest, mature secondary forest, and undisturbed mature plantations) open and disturbed forests in the Western Ghats, India. They used multivariate analysis along with satellite images and found a positive correlation between water quality and forest cover at a localized level. A decrease in forest cover led to an increase in turbidity, total suspended solids (TSSs), and increased number of *Escherichia coli*. Moreover, old forests were more effective in reducing turbidity and TSS than disturbed forests (Singh & Mishra, 2014; Yu et al., 2019) as old-growth forests have a vegetative design that minimizes soil erosion and sediment runoff while promoting soils that provide natural filtration. Therefore, the conservation of old/mature forests is an essential factor for the stabilization of water yield and quality. Ecohydrological interactions are moderated by different factors like forest type, age, climatic zones, soil, scale, and management practices, therefore it is imperative to understand the ecological implication of such drivers on forest–water–climate interactions.

2.3 | Contrasting theories

A multitude of studies have shed light on the varying impacts of changing forest cover on water availability resulting in a range of theories with often different and sometimes contradictory outcomes. Studies comparing less forested and heavily forested catchments indicate that a good forest cover reduces runoff and vice-versa. This is generally attributed to the fact that forests absorb precipitation along their catchment; maintain microclimates through evapotranspiration, canopy interception, groundwater linkages; and regulate streamflow (Jackson et al., 2009; Li et al., 2017; Sahin & Hall, 1996; van Dijk et al., 2012; Wei et al., 2018). At a landscape level, forests influence soil formation and functioning which impacts forest hydrology through water retention, infiltration, percolation, soil moisture storage, release, erosion, and as a medium for roots (Brantley et al., 2017), which forms the basis of sponge theory. "Sponge" theory emphasizes the role of soil profile, texture effect, and soil organic matter content, and spatial variation for water availability within the reach of root systems; categorized under the "sponge" effect of the tree roots, soil and forest litter (Myers, 1983; Spears, 1982). Soil-moisture storage is an important source of water during scarce water periods as the uptake of water by roots in wet periods is slowly released in the dry season for sustained water supply (Bruijnzeel, 2004). Furthermore, drought hit forests also tend to develop deep roots for storage and access to soil and ground water for transpiration (Teuling et al., 2010).

A different school of thought explores the trade-off theory that suggests higher tree densities may result in excessive water loss by transpiration and lower water yield (Calder, 2007) considering trees as water consumers rather than water producers (Ellison et al., 2012). However, the generality of trade-off theory is questioned as it undermines a variety of services such as enhanced water quality, moderation of most peak flows, carbon sequestration, moisture recycling for multiple precipitation cycles, and groundwater recharge generated through forest cover (Ellison et al., 2017; Ilstedt et al., 2016). Deforestation impacts on streamflow are highly influenced by soil type, depth, and terrain features that are often overlooked (Brantley et al., 2017; Ellison et al., 2012) by the supporters of "demand-side/consumer side" of forests.

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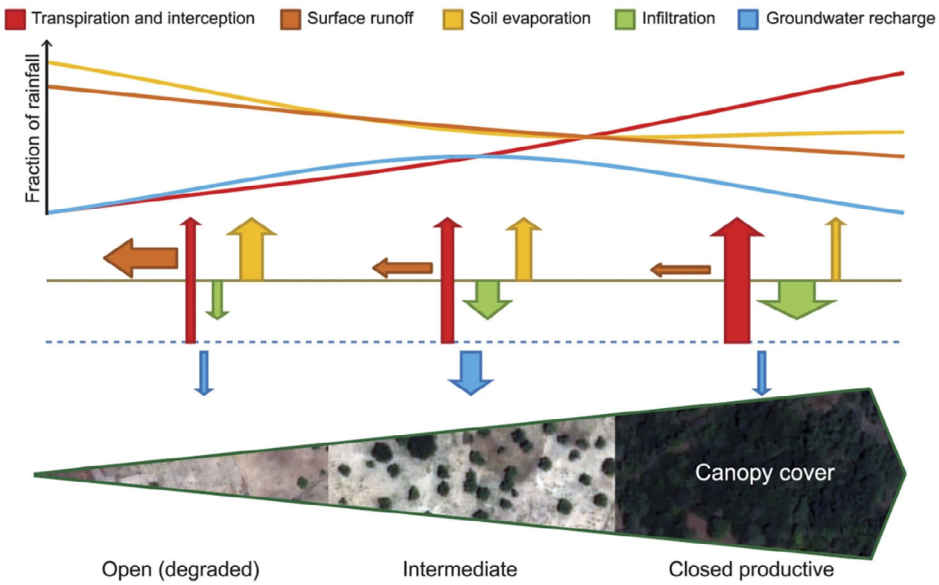

FIGURE 1 Optimum tree cover theory: at an intermediate canopy cover, low surface runoff, and evaporation as well as intermediate transpiration optimize groundwater recharge (*source*: Ilstedt et al., 2016)

On the contrary, studies indicate that soil hydraulic conductivity is enhanced by an increase in tree cover, which in turn regulates the overland flow of water (Bruijnzeel, 2004; van Dijk et al., 2012).

A relatively new theory of optimum tree cover suggests that groundwater recharge maximizes at a specific non-zero tree cover. Increasing tree numbers up to the optimal tree density counterbalances their extra water use, while the opposite happens by exceeding this optimal value (see Figure 1). This theory was tested under specific seasonally dry tropical conditions with the optimum tree cover depending upon the species, climatic conditions, and edaphic factors exclusive to the area (Ilstedt et al., 2016). Field observations suggest that under the above conditions, a maximum amount of rainfall percolates around the edge of a canopy while less water is available for groundwater recharge in areas away from trees, proving negligible groundwater recharge in absence of trees. There is quite a large variation concerning how vegetative assemblages under different bioclimatic scenarios interact with different forms of water production such as runoff, groundwater, and streamflow, which highlights the need for further field studies.

2.4 | Changing land use patterns and eco-hydrology

Aquatic ecosystem services and forests are strongly interconnected through socio-ecological components (Creed et al., 2016), but overexploitation has severely impacted their functioning. Net global forest cover has declined by over 178 million hectares since 1990 (FAO, 2020). Large swathes of forest have been lost to pastures, agricultural land, row crops, urban development, resource extraction, and other land use pressures (Jackson et al., 2009; van Dijk et al., 2012). Such changes in land use patterns affecting forest cover, positive or negative (Calder, 1996), influence hydrology primarily through cumulative changes in hydrological systems. These have compounded effects on base flow, subsurface flows, stormflow, and groundwater levels, leading to substantial changes in both water quantity and quality (Neary, 2016; Vose et al., 2012). Paired watershed studies have found that deforestation can alter base flows and amplify peak flows while reforestation can do the opposite, stabilizing base flows and reducing peak flows (Li et al., 2017). Water stress and scarcity caused by land use changes, extreme deforestation and turbulence in climate patterns further exacerbate these issues.

Another common approach to study the impact of land use and land cover change on runoff-rainfall relationships is through satellite imagery (Arnold et al., 2020; Khare et al., 2015; Seitz et al., 2012). Image analysis has been used in the Athabasca River basin, Canada by comparing historical satellite images with the current land use and correlating it with the river's water quality and quantity (Seitz et al., 2012). The water quality in the basin had significantly deteriorated with an increase in forest removal driven by industrial intensification and water abstraction. The study stressed

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the need for improving monitoring strategies to effectively manage river systems. In the Middle East, a long-term trend analysis over a 30-year period was used to study the impacts of cloud forest on water yield of the Dhofar mountains, which showed an increased and stable forest cover was positively correlated with a higher hydrological regime (Arnold et al., 2020). This improved groundwater recharge was attributed to enhanced cloud interception by forests. Similarly, the impact of natural regeneration of forest cover in Tropical montane cloud forests of Paraibuna river contributed to increased water yield, when the forest cover improvement was reported at 20%; however, in case of cloud forest cover increase lower than 20%, the hydrological regime was moderated by hydro-climatic factors (air temperature and rainfall) (Teixeira et al., 2021). In Europe, hydrological response to land use changes has been carried out via empirical rainfall-runoff modeling using Soil Conservation Service Curve Number (SCS-CN) method (Okoński, 2007). In India, a similar study conducted for a sub-basin of Narmada using GIS (image analysis) and SCS-CN method estimated the runoff to increase by 5% during the time span of 10 years due to land use changes (Khare et al., 2015). Assuming no changes in rainfall patterns occur over this period, the study predicted a two-fold increase in runoff (10%).

These studies imply that forests are prime regulators of a stable hydrological regime and therefore, these interactions should be studied in tandem with the impact, presence and considerations of climatic parameters as highlighted by Black (1998). He further reiterated the need for evaluation of hydrological responses to heterogeneous forests, development of reliable models and statistical tests for assessment, the importance of scaling watersheds for the development of management strategies, the need for understanding inter-relationships between terrestrial, aquatic, and riparian systems to better assess hydrological processes, water ecosystem services, role of global water, runoff and water quality characteristics in tandem with socio-economic ethos. Many of the above have been addressed through continuous efforts of the scientific community with particular strides toward the use of comprehensive datasets for hydrological computation in improved models for projecting impacts of climate change, agricultural activities, and anthropogenic interference on different qualitative and quantitative variables related to water (Arnold et al., 2020; Wei et al., 2018; Yu et al., 2019). Thus, inclusive models focusing on the positive influence of forests on water are required for integrated policy formulation in managing forest and water synergistically and effectively.

3 | WATER AND CLIMATE CHANGE

Water plays an important role in regulating Earth's climate and environment but changing climatic patterns have the potential to, directly and indirectly, influence water resources by altering the hydrological cycle, which in turn, can further accelerate climate risks and impact water quantity. As a result, increasing average water vapor and evaporation rates has been observed to modify precipitation patterns and intensity, resulting in severe extreme events such as flash floods and droughts (Bates et al., 2008; Trenberth, 2005). Almost 18% of daily precipitation extremes (moderate) are attributable to the current warming of 0.85°C, which will increase by 40% in a 2°C warming scenario (Fischer & Knutti, 2015). Extreme weather events are influenced by numerous factors, for example, extreme precipitation does not always lead to floods as there are other factors such as catchment characteristics, catchment sensitivity to flood response, storm magnitude, urbanization, soil moisture, precipitation patterns and duration which determine the flood response (Sharma et al., 2018). There is an overall consensus that the observed warming trend has resulted in large-scale hydrological changes proving that water is intermittently linked to climate. The impact of changing climate on hydrological functions has been a key feature in IPCC's assessment, as an understanding of these responses is crucial for the management of water resources in the changing climatic regime, which would further help in the development of adaptation and mitigation strategies.

Changing climate and its associated impacts on different components of the hydrological cycle have been studied through hydrological and statistical modeling approaches. Studies on the latter, typically analyze historical data sets for trend analysis and assess the impact of climate change on groundwater, streamflow/discharge/runoff, and extreme events (Booij et al., 2019; Earman & Dettinger, 2011; Kumar et al., 2019; Laloo et al., 2020; Mahmood & Jia, 2019; Rao et al., 2020; Thomas et al., 2015; Trisurat et al., 2016; Zhang et al., 2015; Zhou et al., 2011). Hydrological modeling is an extended method to study the impact of climatic variability on water. Studies commonly use models like soil and water assessment tool (SWAT), MODFLOW, water evaluation and planning system (WEAP), MIKE SHE with different future climatic scenarios to predict water quantity for various moderate and extreme climatic scenarios (Bhagwat & Maity, 2014; Im et al., 2009; Li et al., 2015; Mahmood & Jia, 2019; Rickards et al., 2020; Kumar, 2012; Wang et al., 2013; Yilmaz, 2015). However, most of these models have a specific catchment focus, and thus are unable to address atmospheric moisture flows, cross-continental transport of water vapor across terrestrial surfaces which influences

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precipitation recycling and water availability in downwind locations (Ellison et al., 2012). These limitations lead to underutilization and misrepresentation of forests potential for an improved hydrological regime. Adequate "prediction of water quantity" should also focus on understanding and estimation of the linkages between these partitioned components of hydrological cycle.

4 | IMPACT OF CLIMATE CHANGE ON STREAMFLOW, GROUNDWATER, EXTREME EVENTS: STATISTICAL, AND HYDROLOGICAL MODELING APPROACH

The streamflow in a river system is dependent on the rainfall patterns viz., its spatial and temporal distribution, its intensity and duration among many other physical and climatic factors. Streamflow is also an indicator to measure the quantity of water inflow to a waterbody and has been used extensively in research for quantitative and qualitative analysis for different purposes (Marshall & Randhir, 2008; Somorowska, 2017; Zhang et al., 2017). The impacts of climate change on streamflow, groundwater, and extreme events are studied typically through trend analysis, computing rainfall-runoff relationship, and other empirical methods. Climate change impacts on runoff, soil moisture and evapotranspiration by comparing water fluxes through the rainfall-runoff model (MODHYDROLOG) have been assessed for 28 Australian catchments (Chiew & McMahon, 2002). The incorporation of global climate models (GCMs) for future predictions indicates severe runoff modifications in wet and temperate catchments as well as ephemeral catchments due to change in precipitation patterns. Soil moisture levels for drier catchments were found to be more sensitive as compared to temperate catchments. Keeping aside the large uncertainties in the estimated values, the study suggested the need to account for these impacts in any kind of planning process (Chiew & McMahon, 2002). Impact of climate change on seasonal water resources of Thailand was assessed with HEC-HMS model which incorporated land use, soil properties and simulated water availability using precipitation data for SRES A2 and B2 scenarios from GCMs. Increased dry season and flow were observed at the national level relative to the baseline period involving uncertainty (Deb et al., 2018). The outcomes of such studies are helpful in formulation of better water management strategies to minimize impacts of potential extreme events. Although, the GCMs are widely used for predictions and potential implications of climate change, the over estimation of scenario forecasts is largely impacted by large physical uncertainties (Pitman et al., 2009) due to the improper understanding and modeling of water vapor feedback and the cloud system. In light of this, it is suggested that CMIP6 GCMs with high equilibrium climate sensitivity should be avoided for policy formulations (Scafetta, 2021).

Mann–Kendall test (MKT) and change-point analysis have been used extensively to determine significant trends along with pre and post periods (Kendall, 1948; Mann, 1945). A study on the Yi River basin in China analyzed long-term trends of precipitation and streamflow using MKT, defining the change point year to divide the time series into two time periods. The study used climate elasticity and linear regression techniques to distinguish between the effect of climate change and anthropogenic interference on annual runoff with higher influence being attributed to precipitation variability (Zhang et al., 2017). Similarly, a study conducted for Lasica catchment in Poland used MKT and Cumulative Sum (CUSUM) test for long-term trend analysis of precipitation, air temperature, potential evapotranspiration, streamflow, and groundwater level and found a significant effect in the form of reduction in groundwater recharge, increased evapotranspiration which in turn resulted in reduced runoff (Somorowska, 2017). Average annual precipitation, intensity, air temperature and seasonal air temperature were identified as the major factors influencing runoff using correlation, residual error, and principal component analysis in a study for North Korea (Ri et al., 2019). The study developed an empirical hydrological model to assess the runoff patterns and tested the same for the Tumen River basin for water resource computation. Although the model would benefit in mapping water resources for better management of the same and can assess climate change impacts, it does not consider surface-runoff infiltration and runoff process (Ri et al., 2019). Spatio-temporal analysis for rainfall extremes of Nagavali and Vamsadhara basins of Eastern India have also been conducted for three different time periods using MKT and Sen-slope method (Rao et al., 2020). The findings report a decreasing trend of rainfall extremes except in the monsoon season, which had an increasing trend. In another study on the Narmada basin, Central India, trend analysis for monsoon rainfall variability for the entire basin revealed that high intensity storms and rainfall were more prominent in the recent two decades with an increasing trend of extreme events in the basin (Thomas et al., 2015). The study highlighted the need for better assessment of the basin through dynamic modeling for management of water resources and the development of mitigation strategies. Trend analysis of rainfall and correlation with groundwater levels in Kumadvati watershed, Karnataka

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(Southern India) indicated decreasing trends for both rainfall and groundwater; with intensive irrigation activities and the influx of population being cited as a possible reason for groundwater depletion (Laloo et al., 2020). Such studies underline the need to contextualize planning and management of water resources with agriculture and flood management.

Hydrological modeling under different climate scenarios can also provide insights into the impacts of changing climate. The variability in temperature, rainfall, and storm intensity; and its resultant impacts on soil moisture, groundwater and streamflow have been assessed for dry and wet seasons by utilizing long-term data and SWAT in Southern China (Zhou et al., 2011). The study reported an increased susceptibility of the region to extreme weather events based on significant reductions in soil moisture levels coupled with increased streamflow and groundwater levels in the wet season, intensified rainfall storms, and a decrease in the number of rainy days. Gosain et al. (2006) evaluated the impact of changing climate on water resources in Madhya Pradesh using the Regional Climate Model (RCM) and SWAT. The analysis indicated large variations at the district level due to regional differences in water demand for agriculture and adaptive capacity. Impact assessment of climate change on water resources using trend analysis for the temperature, rainfall, and groundwater levels was conducted for the entire Narmada basin (Sudheer, 2014). Furthermore, the SWAT model (inputs: digital elevation model, land use, soil type, and climatic datasets) and RCM for different Representative Concentration Pathways (RCP) were assessed for future climatic scenarios on hydrological variables (runoff magnitude and pattern). The magnitude of rainfall patterns, variability in temperature had an increasing trend while extreme flood events had a mixed trend. Such integrated studies on climate change impact assessment for major river basins incorporating topography, soil, and land use/land cover are requisite for the formulation of region-specific adaptation strategies.

There is no "best" method to quantify the hydro-meteorological relationship as both statistical and hydrological models come with their own sets of advantages and drawbacks. Statistical models are easier to use and demand only basic meteorological datasets (temperature and precipitation) and hydrological datasets (runoff series and discharge) but lack the details of hydrological modeling. Hydrological models incorporate different components and their interactions but require extensive datasets and validation, which is extremely challenging. The incorporation of GCMs and downscaling for future predictions involves accepting uncertainty in the results. Therefore, many studies have specified the inclusion of both methods to yield better predictions for the management of water resources. Hence, existing eco-hydrological studies focus on spatio-temporal analysis of climatic variability on watershed services and functions with very few of them focusing on all three aspects of climate change, water, and forest (Sudheer, 2014; Thomas et al., 2015; Wang et al., 2011).

5 | INTERLINKAGES BETWEEN FORESTS, WATER, AND CLIMATE CHANGE: PRESENT UNDERSTANDING OF THE LINKAGES

Forests act as both moderators and catalysts in many environmental interactions (Ford et al., 2011; Seidl et al., 2017; Vose et al., 2011). The ecology, distribution, biomes, and structure of global vegetation patterns are strongly influenced by climate with an estimated one-third of global forests under the threat of composition shift due to climate change (Dale et al., 2010; Shriner et al., 1998). Forests of different climatic zones, types, ages, regions react in a diversified manner to climatic variability due to differences in sensitivity and resilience to disturbances (Bonan, 2008; Canadell & Raupach, 2008; Devi et al., 2018; Goldstein et al., 2000; Patasaraiya et al., 2021; Ravindranath et al., 2006; Yu et al., 2019). These sensitivities of forest ecosystems are further impacted by water availability. Even though a significant part (one-third) of Earth's land is too dry to support tree growth, climatic variability (increased temperature, change in precipitation patterns and atmospheric compositions) aids forest growth in the presence of abundant water (Allen, 2009; Boisvenue & Running, 2006). The influence of climate on forests and a hugely debated role of current afforestation practices to combat climate crisis (Di Sacco et al., 2021) implies the complicated interaction of natural phenomena which transcends multi-disciplinary fields forms a complex nexus. The understanding of synergies and trade-offs between forests, water, and climate change will provide a more comprehensive approach to the understanding of the nexus.

There has been some exploration of the cumulative effect of forests and climate change on hydrological processes through paired watershed studies (Yurtseven et al., 2018), landscape level monitoring (gauge station data, long-term trend analysis), and a combination of both methods (Neary, 2016). Yurtseven et al. (2018) conducted an insightful experimental study assessing the impact of different forest management practices on runoff generation using paired

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watershed treatment in Turkey. Thinning treatments were carried at varying intensities over different time periods which resulted in slightly increased runoff suggesting light thinning treatments could work for increasing the runoff statistically, provided there is an adequate amount of precipitation. Therefore, forest management techniques may play a prominent role in managing watersheds and should be developed based on the regional climatic zones and precipitation patterns. At a global scale, the change in vegetation cover and climate on annual runoff was assessed, 30.7 ± 22.5% of which was attributed to change in vegetation cover while the rest was attributed to climate change. As a result, inclusion of vegetation cover change in global models for determining the changes in global water resource due to climate change in forest-dominant areas assumes significance (Wei et al., 2018). Duan et al. (2017) quantified the impact of climate variability, forest cover change, and permafrost thaw on streamflow in Northern China using forest cover data and Normalized Difference Vegetation Index (NDVI) to assess changes in forest dynamics, temperature, precipitation, and streamflow. Trend analysis and sensitivity methods were used for estimation of the climatic variability and forest cover change on streamflow. They found forest cover variability to be the dominant driver in influencing annual streamflow whereas climatic variability as a secondary factor. Such an approach can help identify areas of focus for various natural resource management plans. SWAT model has also been extensively used for quantifying the effects of climate and forest change on water. Analysis of river discharge through SWAT and GCMs under different RCP scenarios incorporating climate, soil, forest, land use, and runoff data are common with results generally correlating decreased forest biomass with increased annual runoffs. Such results are consistent with other GCM projections that indicate increasing annual temperatures and precipitation have exacerbated impacts on water resources and vegetation (Yu et al., 2020).

Climate and forest related drivers require particular attention to better assess impacts on water quantity, which would be pivotal for future forest and water ecosystem management (Duan et al., 2017; Neary, 2016). Studies in India have been sparse and unconsolidated. Rangan et al. (2010) assessed regional rainfall patterns and the potential impact of wattle (Acacia mearnsii) plantations on water availability in Palnis Hills, Tamil Nadu. They found changing rainfall patterns to be more influential in determining regional water availability than the wattle plantations, which was further supported by the perception of the locals. Such findings underline the need for comparative studies between ecological changes, regional economic fluctuations, and rainfall patterns. Singh and Mishra (2012) assessed the effect of forest cover on water yield in Peninsular Western Ghats by using forest cover and land use-land cover maps with the help of the water budget equation for estimating the monthly runoff. Using rainfall as the principal climatic parameter, they estimated runoff by correlating water yield with forest type (disturbed, primary, and mature forests). Similarly, there have been a few studies with a focus on using land use change and climate change scenarios to project water budgets (Kundu et al., 2017; Wilk & Hughes, 2002). However, there is a dearth of meaningful integration of forest ecology under changing climate scenarios and its consequences on water availability in the Indian context. The prominent reason for this gap is the convoluted and complex interactions of these three components, data paucity, and uncertainty in existing models and climate change scenarios. Furthermore, experimental studies to validate such projections of water quantity are seldom time and cost-effective. The sustainable and improved management of forest and water resources in increasingly disturbed climate scenarios requires adhesion of policies to the interlinkages between these phenomena and also requires exploration of different aspects of livelihood, human interferences, changes, consumption patterns which could be facilitated by a further detailed exploration of such interactions.

6 | CONCLUSION

It is certain that forests, climate, and water are interlinked and changes in one affect the others. There is a wide array of synergies and trade-offs amongst their interactions, which also permeates down to analytical studies with an attempt to model their impacts on each other. The role of forests in sustaining water resources has been undeniably cemented through studies on the impact of deforestation and afforestation on streamflow, stormflow, groundwater recharge, and land use land change scenarios (Yu et al., 2019) and their regulation of water flow through structural properties (Neary et al., 2009). However, a lack of coherence due to contrasting theories and divided views on the role of forests and plantations for a healthy hydrological regulation, further compounded by the overlooking of climatic parameters has been a limiting factor for a more holistic understanding of the issue. The intricate mechanisms by which forest catchments are influenced by climatic changes such as evapotranspiration and flux-regulation of atmospheric moisture determining rainfall patterns have also been readily established (Ellison et al., 2017). Similarly, changing forest cover and climate variability have been identified as major drivers for inducing hydrological modifications in a forest-dominated land, as

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changing forest and climate have "offsetting and additive effects" on runoff (Li et al., 2017) complicating the prediction and management of water resources (Wei et al., 2018). Furthermore, there is a strong correlation between streamflow response and forest cover with edaphic, topographical, and climatic factors being key influencers (Hibbert, 1965). Climate, in particular, has been widely documented to influence annual runoff along with forest cover change at multiple spatial scales (Duan et al., 2017; Li et al., 2017; Zhang et al., 2017). Anthropogenic interferences alter each sector both individually and synergistically adding to the complexity and entailing a more nuanced understanding of synergies and trade-offs between forest, water, and climate change for designing effective policies and strategies in managing water and forests in context to climate change. The water-energy-food nexus has already begun to garner the requisite attention to these interlinkages for better addressing food and water security (Endo et al., 2017; Tidwell, 2016). Similarly, there is a pressing need to incorporate forests, particularly Forest and Landscape Restoration (FLR) approaches, into this nexus (Melo et al., 2020). There is also a need to include regional rainfall variability and climatic patterns for contextual assessments that would help develop adaptive resilience-based strategies for water and land resource management (Rangan et al., 2010). Current research has probed toward better insight into the nexus through empirical and statistical models. However, prevalent uncertainties in their results, as well as the fact that they are built upon a set of localized assumptions unique to different approaches and scenarios, undeniably stress the need for further exploration (Pitman et al., 2009). Therefore, there is a need to study these components in an integrated manner rather than in silos to accommodate these interactions in the planning and management of forests and water.

AUTHOR CONTRIBUTIONS

Manisha Singh: Conceptualization (equal); writing – original draft (lead); writing – review and editing (equal). **Bhaskar Sinha:** Conceptualization (lead); supervision (lead); writing – review and editing (equal). **Jigyasa Bisaria:** Conceptualization (equal); supervision (equal); visualization (equal). **Thomas Thomas:** Conceptualization (equal); supervision (equal); writing – review and editing (equal). **Pankaj Srivastava:** Conceptualization (equal); writing – review and editing (equal).

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CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

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